

Nikolaos Xiros

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About

As a Associate Researcher at the Athena Research Center and a MEng graduate in Electrical and Computer Engineering from the National Technical University of Athens, I'm passionate about Machine Learning and Natural Language Processing. My research interests involve investigating the reasoning capabilities of NLP models, designing interpretability algorithms, and enhancing the safety and reliability of AI systems. Beyond technical development, I am dedicated to analyzing how AI interacts with society by evaluating fairness, transparency, and ethical implications in real-world deployment scenarios.

Education

- **National Technical University of Athens** Oct 2020 – Oct 2025
Integrated Master's (MEng) in Electrical and Computer Engineering Athens, Greece
 - GPA: 9.08/10 (High Distinction – Ranked top 7% of class)
 - Thesis: *Refining CLIP Visual and Language Representations via Self-Supervised Visual Structure*
 - Supervisor: **Prof. Alexandros Potamianos**
- **Technical University of Denmark (DTU)** Aug 2023 – Jan 2024
Erasmus+ Exchange Program Lyngby, Denmark

Research Experience

- **Institute for Language and Speech Processing (ILSP)** Oct 2025 – Present
Associate Researcher (previously Research Intern) Athens, Greece
 - Investigated novel architectures for Mixture of Experts (MoE) models and developed a novel router paradigm with enhanced explainability on the expert selection process, results be submitted to **EMNLP 2026**.
 - Explored the faithfulness of LLMs when encountering unsolvable math problems, results to be submitted to **EACL 2027**.
 - Developing and training mechanistic interpretability frameworks (Sparse Autoencoders (SAEs) and Transcoders), extended to multimodal architectures.
 - Pre-training, fine-tuning, evaluating, and deploying large-scale ML models on HPC infrastructure.
- **University of Science and Technology of China (USTC)** Jul 2025 – Aug 2025
Future Scientist Exchange Program (FuSEP) – Summer Research Hefei, China
 - Participated in a 5-week international research program at USTC, collaborating with leading researchers.
 - Worked on data-driven and neural network-based modeling to capture and control the dynamics of a novel soft robotic system.

Skills

- **Programming:** Python, C/C++, SQL, Web Development, Bash, Git, Operating Systems
- **Machine Learning:** PyTorch, HuggingFace, NumPy, OpenCV, Megatron-LM, Deepspeed

Projects

- **IEEE NTUA Student Branch** Sep 2022 – Sep 2024
Member Athens, Greece
 - Active volunteer and project contributor in the IEEE student branch at NTUA.
 - Participated in the Bosch Future Mobility Challenge, developing robotics and AI modules for autonomous driving tasks.

Honours and Awards

- Mathematics Competitions – Hellenic Mathematical Society (Thales & Euclid distinctions)
- Physics Competition – Union of Greek Physicists (6th Nationwide, 30th National Competition)
- Graduation with High Distinction (GPA 9.08/10) – NTUA ECE

Languages

Greek: Native

English: C2 (Proficient) – Listening, Reading, Speaking, Writing

Publications

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [C.1] Alexis Filippakopoulos, **Nikolaos Xiros**, Elias Kallioras, Efthymios Georgiou, Alexandros Potamianos. (2025). **Segregate, Refine, Integrate: Decomposing Multimodal Fusion for Sentiment Analysis**. Accepted in *Interspeech 2026*.
- [S.1] **Nikolaos Xiros**, Georgios Paraskevopoulos. (2026). **Beyond Magnitude: Contrastive Routing for Modular Mixture-of-Experts**. Manuscript submitted for publication in *EMNLP 2026*.
- [S.2] **Nikolaos Xiros**, Maria-Eleni Zoumpoulidi, Georgios Paraskevopoulos. (2026). **Knowledge Knows, Verbalization Tells: Disentangling Latent Directions for Mathematical Solvability in LLMs**. Manuscript to be submitted for publication in *EACL 2027*. Preprint available on *arXiv*.